

EcoNova AI

Team Name: EcoNovaX

Team Leader Name: Bharti Chavda

College Name: Saurashtra University

PROBLEM STATEMENT

Urban waste management systems are inefficient, slow, and heavily dependent on manual processes. Cities generate massive amounts of waste daily, but lack intelligent systems to classify, prioritize, and resolve complaints in real time.

This results in delayed response, duplicate complaints, poor resource utilization, and increasing environmental pollution. A smart AI-powered solution is needed to automate waste detection, complaint handling, and optimized municipal decision-making.



ABSTRACT

EcoNova AI is an AI-powered waste management intelligence system that transforms citizen-reported waste images into actionable municipal tasks in seconds.

Using YOLOv8-based image classification, duplicate detection, and priority scoring, the system automatically analyzes waste complaints, assigns urgency, and updates a real-time municipal dashboard.

The goal is to improve waste collection efficiency, reduce manual workload, and support sustainable urban development.

AI-Powered Waste Management System
Seamless Cloud Connectivity for Smarter Cities.



TECH STACK



Backend: Python, Django, Django REST Framework

Database: MySQL

AI Engine: YOLOv8, OpenCV

Frontend: HTML, CSS, Bootstrap, JavaScript



EXISTING SOLUTIONS & DRAWBACKS

Existing System:

- Manual complaint handling by municipal staff
- Fixed waste collection schedules
- No AI-based classification
- First-come-first-served resolution

Drawbacks:

- Slow complaint resolution
- Duplicate complaint overload
- Inefficient resource usage
- No priority system for urgent issues
- High operational cost and delay



Paper complaints.
Slow processes. Frustrated
municipal workers.

PROPOSED SOLUTION



EcoNova AI automates the entire waste management pipeline:

1. Citizen uploads waste image
2. AI detects and classifies waste type
3. System checks duplicate complaints
4. Priority score is assigned
5. Complaint is sent to municipal dashboard
6. Field teams receive optimized tasks

This reduces manual intervention and speeds up resolution from days to minutes.



AI-POWERED WASTE MANAGEMENT SYSTEM

Smart City. Clean Future.

1 CAPTURE

Citizen captures waste image



2 AI ANALYSIS

AI analyzes image, detects waste type and severity



AI DETECTION RESULTS

Waste Type : Mixed Waste
Severity : High
Location : Sector 4S, City
Time : 10:30 AM



3 CLOUD PLATFORM

Data is uploaded to cloud and processed



LOCATION
MAPPING



DATA
ANALYTICS



ALERT
SYSTEM

4 SMART ACTION

Auto-notify nearest cleaning team & track resolution

DASHBOARD

NEW COMPLAINTS

23

High Priority

RESOLVED

76%

This Week

ACTIVE CLEANING UNITS

12

Units Deployed



AVG. RESPONSE TIME

28 mins



5 RESOLUTION

Waste collected & location updated in real-time



BENEFITS



Faster Response



Cleaner Cities



Data-Driven
Decisions



Efficient Resource
Utilization



Better Quality
of Life

ARCHITECTURE / WORKFLOW

Workflow:

Citizen → Upload Photo → AI Engine (YOLOv8) → Classification + Duplicate Detection → Priority Scoring
→ MySQL Database → Municipal Dashboard → Resolution Team Dispatch

Key Features:

- Real-time AI processing
- Centralized dashboard
- Instant complaint triage
- Smart data storage

UNIQUENESS / INNOVATION

1. AI-based waste classification (6 categories)
2. Duplicate complaint detection system
3. AI-generated priority scoring system
4. Eco Score gamification for citizens
5. Real-time municipal dashboard
6. Future-ready IoT integration support



ADVANTAGES & LIMITATIONS

Advantages:

1. Faster complaint resolution
2. Reduced manual workload
3. Better recycling accuracy
4. Cost and fuel optimization
5. Improved citizen participation
6. Smart city compatibility

Limitations:

1. Requires image quality for accurate detection
2. Needs training data for regional waste types
3. Internet dependency for reporting
4. Initial setup cost for municipalities

CONCLUSION



EcoNova AI provides a scalable and intelligent solution for modern urban waste management.

By combining computer vision, automation, and smart prioritization, it significantly improves efficiency, reduces delays, and promotes cleaner cities.

It supports sustainable development goals and can be expanded into a full smart city waste intelligence system.



REFERENCES



1. YOLOv8 Documentation (Ultralytics)
 2. OpenCV Official Documentation
 3. Django Framework Documentation
 4. Government of India Swachh Bharat Mission Reports
 5. Smart City Mission Guidelines
 6. Research papers on AI-based waste classification systems
- 